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RESEARCH-ARTICLE

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ECG Signal Denoising based on 2D-LMS Filter in Fractional Fourier Domain

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ABSTRACT

ECG is a non-stationary electrical signal, and the process of collecting ECG signal inevitably contains various noises. This paper proposed a new method for ECG signal denoising in fractional Fourier domain. Firstly transform the signal into fractional Fourier domain, and then use 2D-LMS filter to process the characteristics of signal, finally get the denoised signal in time domain by inverse transformation. Simulation experiments are carried out with the data of MIT-BIH database. The results show that this method can restore the waveform characteristics of ECG signal, effectively improve the SNR of noisy ECG signal and reduce its MSE.

CCS CONCEPTS

• **Hardware**; • **Communication hardware, interfaces and storage**; • **Signal processing systems**; • **Noise reduction**; • **Applied computing**; • **Life and medical sciences**; • **Health informatics**;

KEYWORDS

ECG signal, ECG denoising, LMS filter, fractional Fourier domain, SNR

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1 INTRODUCTION

Electrocardiogram (ECG) is a non-stationary electrical signal that reflects the biosignal activity of the heart. It is collected through the contact between electrodes and skin surface. During the collection process, various noises such as baseline wander (BW), muscle artifact (MA), and electrode motion artifact (EM) will be generated, due to the influence of acquisition equipment, electromagnetic environment, etc. [1]. On the one hand, these noises will affect the waveform and amplitude of the collected ECG signal, resulting in the inability to distinguish the regions and intervals of the ECG

single beat. On the other hand, many studies on ECG signal are to detect abnormal parts in ECG signal, which contain pathological information of cardiovascular diseases, and these noises will seriously affect disease diagnosis based on ECG signal. Therefore, denoising the collected ECG signal is an indispensable and important part for the analysis of ECG.

The methods of ECG signal denoising mostly use the characteristics of the signal in frequency domain, including methods based on threshold [2–4], filtering [5–7], morphology [8], empirical mode decomposition (EMD) [9–11] and transform [12–14]. Both threshold-based and filtering-based methods select the specific frequencies of the signal for reservation. The morphology-based method describes the basic characteristics of the image formed by ECG, and then denoise the noise, while EMD-based method is an adaptive analysis method using Hilbert-Huang transform. The transform-based method also makes use of the time-frequency characteristics of ECG signal, and the commonly used transforms include Fourier transform (FT) and wavelet transform. The difference of this paper is that it processes the ECG signal in transform domain, and then restores it through the inverse transform. It can reduce the interference of noise and retain the original characteristics of ECG signal. In addition, there are also many researches specifically aimed at the three main noises (BW/MA/EM) of ECG signal [15–18].

The fractional Fourier transform (FRFT) is a generalization of the FT. It has one more degree of freedom than the FT, then it can rotate the plane of the signal in transform domain. The effect of noise denoising in fractional Fourier domain has more potential than Fourier domain. In view of this, this paper proposed an ECG signal denoising method based on two-dimensional adaptive least mean square error filter (2D-LMS) in fractional Fourier domain. Experiments show that this method can restore the waveform characteristics of ECG signal, effectively improve the signal-to-noise ratio (SNR) of noisy ECG signal and reduce its mean square error (MSE). Its applicability is better than conventional LMS filter, and its denoising effect is better than Wiener filter.

The outline of the paper is as mentioned below. Section 1 introduces the background of ECG signal processing, and explains the research field and purpose of this paper. In Section 2, the main theory of FRFT and filters are discussed. In Section 3, a new filter 2D-LMS that is more suitable for ECG signal in fractional Fourier domain is proposed on the basis of LMS. Section 4 deals with the simulation experiments of data of Noise Stress Test Database and we finally conclude in Section 5.

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2 PRELIMINARIES

2.1 FRFT

The FT is a signal analysis method in frequency domain, and the FRFT is a more generalized Fourier transform, which adds more new advantages while retaining the excellent properties of the FT. The FRFT has one more degree of freedom than the FT, which is the rotation angle. By adjusting the rotation angle, a plane with the largest signal spectrum between time domain and frequency domain can be selected. In this way, the frequency domain characteristics of the signal are the most significant. That is the advantage of the FRFT.

The definition of the FRFT is as follows [19]. For a signal $x(t)$, its FRFT is

$$F_p(u) = \int_{-\infty}^{\infty} K_p(t, u) x(t) dt \quad (1)$$

Where p is the order of the FRFT, and the kernel function $K_p(u, t)$ is expressed as

$$K_p(t, u) = \begin{cases} \sqrt{(1 - j \cot \alpha)} \exp \left[j \pi (u^2 + t^2) \cot \alpha - 2ut \csc \alpha \right], & \alpha \neq n\pi \\ \delta(t - u), & \alpha = 2n\pi \\ \delta(t + u), & \alpha = (2n + 1)\pi \end{cases} \quad (2)$$

Where $\alpha = p\pi/2$, and n is an integer.

When $p = 1$, $F_1(u)$ is the FT, and the relevant proof is in [20]. The selection range of the order p is related to the rotation angle. After the rotation angle is greater than 2π , the characteristics of the signal in fractional Fourier domain will appear repeatedly, so getting the optimal order of the FRFT only needs to consider the values in $p = [0, 4]$, $\alpha = [0, 2\pi]$.

2.2 Filters

2.2.1 Wiener Filter. Wiener filter is a modern digital filter. The traditional filter is only suitable for the situation where the signal and noise do not overlap in frequency bands. When the frequency bands of signal and noise overlap, traditional filters will be powerless, and modern digital filters are needed. Wiener filter makes use the power spectrum and statistical characteristics of the signal and noise (such as autocorrelation function, correlation function). Under certain constraints, the square of the difference between output and the expected output is minimized, so Wiener filter is also called the minimum mean square error filter [21, 22].

For a signal $x(t)$ with a finite length of M , the expected signal is $d(t)$, R_1 is the correlation matrix of $x(t)$ and $d(t)$, R_2 is the autocorrelation matrix of $x(t)$, and the output signal of the Wiener filter is

$$y(k) = \sum_{m=0}^M h(m) x(k - m) = H(m) * X(k) \quad (3)$$

Where $k = [1, M]$ and $H(m)$ is a series of coefficients with length M to minimize the error $e(k) = d(k) - y(k)$

$$H(m) = R_1^{-1} R_2 \quad (4)$$

Although Wiener filter is more suitable for stationary signals, the potential of Wiener filter in denoising of ECG signal can still be tapped due to the periodic characteristics of ECG signal. Certainly, the use of Wiener filter will inevitably affect the high frequency

segment of ECG signal to a certain extent while denoising the noise in the low frequency segment.

2.2.2 LMS Filter. LMS filter is an optimal filtering method developed on the basis of Wiener filter, Kalman filter and other linear filter. It was initially used when the prior knowledge of statistical characteristics of signal and noise was not available. In the application of ECG signal processing, LMS filter has stronger adaptability and better filtering performance [23].

For a signal $x(t)$ with a finite length of M , the expected signal is $d(t)$, and the output signal [24] of the LMS filter is

$$y(k) = W(M) * x(k) \quad (5)$$

Where

$$W(k+1) = W(k) + \lambda e(k) x(k) = W(k) + \lambda [d(k) - y(k)] x(k) \quad (6)$$

And use iterative method until W converges.

3 MAIN METHOD

Through the FRFT, the frequency characteristics of the signal will be clearly displayed in fractional Fourier domain. The noise of the noisy signal does not have exactly the same characteristics as the clean signal, which creates a particularly good condition for signal denoising. The ECG is a signal with a certain periodicity, and its performance in fractional Fourier domain is relatively regular. The noises (BW/MA/EM) of ECG signal are different from the clean signal in fractional Fourier domain. Therefore, it is a very attractive and potential method to denoise the noise of ECG signal through FRFT.

Filter is a frequency selection device that can pass specific frequency components of the signal while greatly attenuating other frequency components. Classical filters include low-pass filters, high-pass filters, band-pass filters, and band-stop filters, which can select frequency segments within a certain range and keep the remaining frequency segments as small as possible. In the processing of ECG signal, the more regular noise is contained in the high-frequency segment, while the white noise is contained in the low-frequency segment. However, simply taking the frequency of certain frequency bands cannot remove the noise contained. Instead, it is likely to erase the original low-frequency characteristics of ECG signal. Therefore, it is particularly critical to select a suitable filter and adapt it to the denoising of ECG signal in fractional Fourier domain.

3.1 2D-LMS Filter

For ECG signal in fractional Fourier domain, the effect of using LMS filter directly will be affected by M in function (5). When M is small or too large, W will diverge, and $y(k)$ will be very large, which is an exponential multiple of $d(k)$. Different from LMS, in view of the fact that ECG signal in fractional Fourier domain is in complex form, an improved 2D-LMS filter is proposed in this paper. For $x(k) = a + bj$, let $x_1(k) = a$, $x_2(k) = b$, and the related formulas become

$$x^*(k) = \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} \quad (7)$$

$$y(k) = W^*(M) * x^*(k) = \begin{bmatrix} W_1(M) & W_2(M) \end{bmatrix} \begin{bmatrix} x_1(k) \\ j * x_2(k) \end{bmatrix} \quad (8)$$

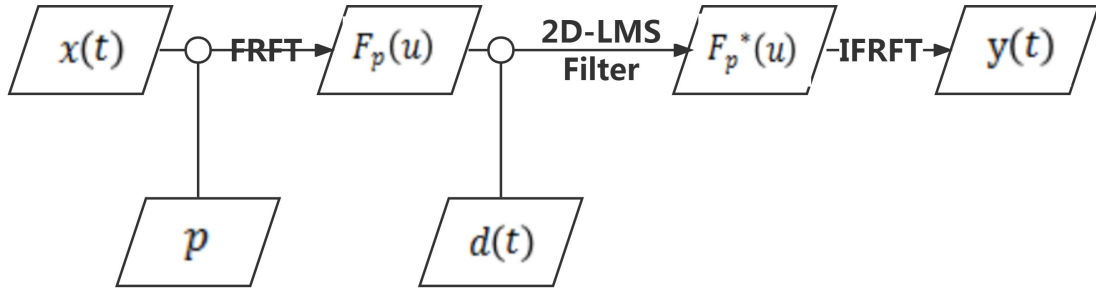


Figure 1: The data flow diagram of ECG signal denoising method based on 2D-LMS filter in fractional Fourier domain

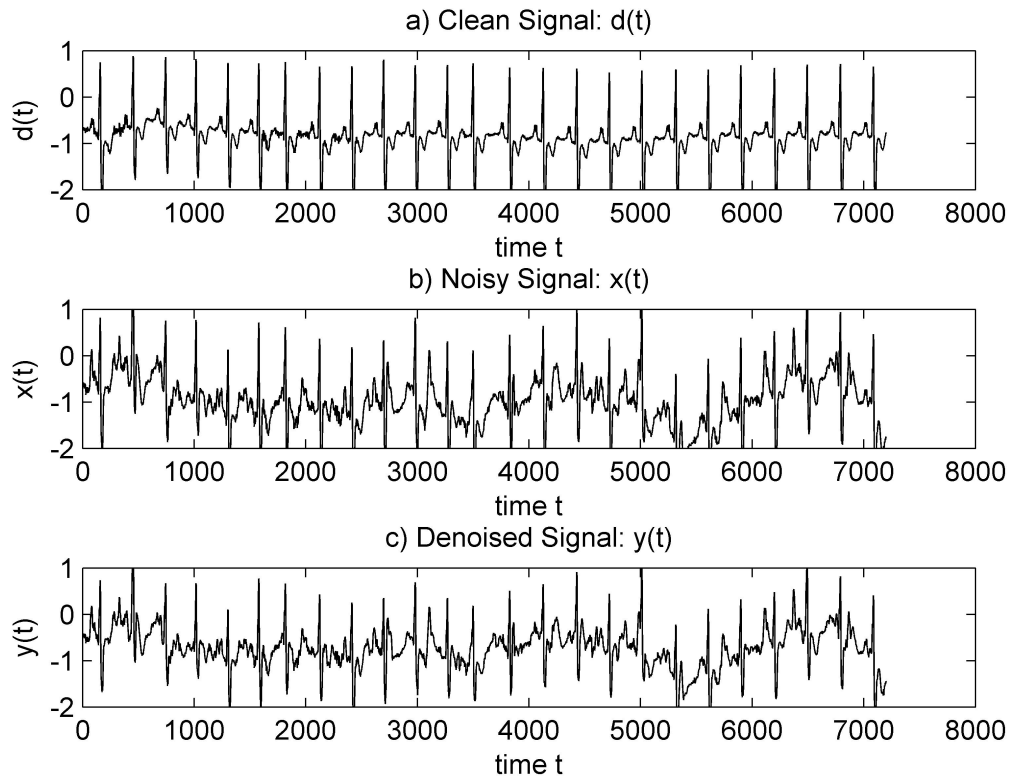


Figure 2: a): The clean ECG signal without noise. b): The noisy ECG signal with noise. c): The denoised ECG signal in time domain.

$$W_1(k+1) = W_1(k) + \lambda e_1(k) x_1(k) = W_1(k) + \lambda [d_1(k) - y_1(k)] x_1(k) \quad (9)$$

Where $d_1(k)$, $y_1(k)$ are similar to $x_1(k)$, $W_2(k)$ and $W_1(k)$ are calculated the same, the details of the specific calculations are not explained here. 2D-LMS filter avoids the divergence of the parameter W , and can be better applied to ECG signal denoising in fractional Fourier domain.

3.2 Main Steps

The main steps of ECG signal denoising method based on 2D-LMS filter in fractional Fourier domain are as follows. Firstly, do a FRFT on the noisy signal $x(t)$, transform the signal from time domain to fractional Fourier domain, and select the most appropriate FRFT order p according to the value of the spectrum in fractional Fourier domain to obtain the p -order FRFT of $x(t)$, namely $F_p(u)$. Secondly,

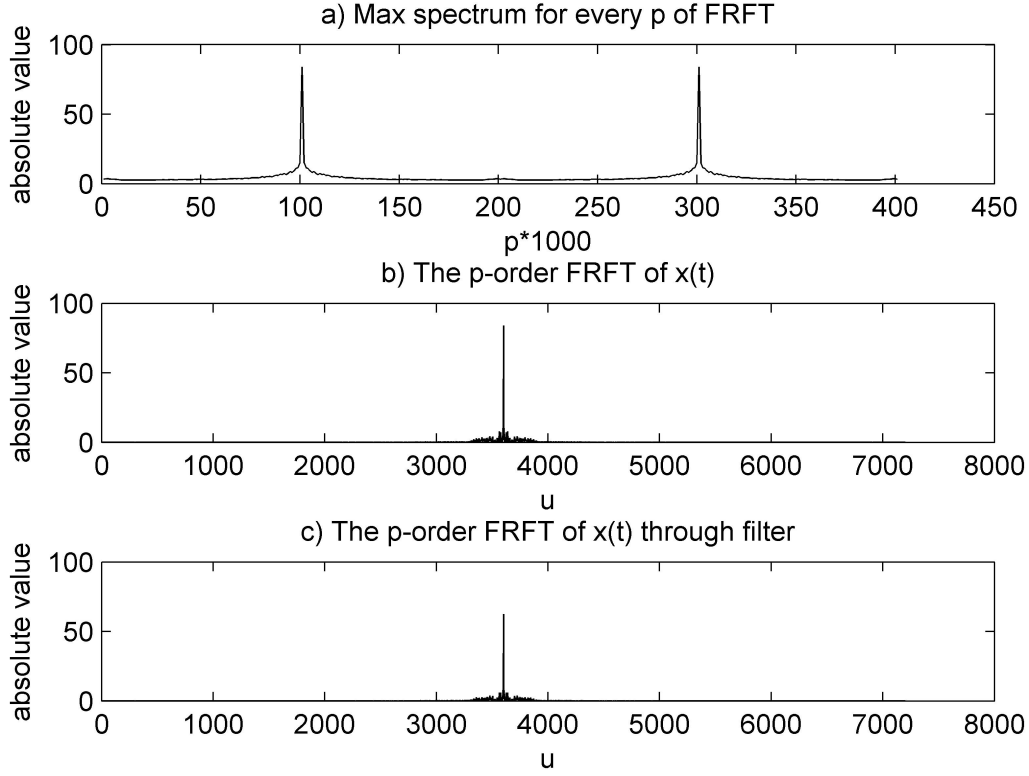


Figure 3: a): The selection process of the best FRFT order. b): The ECG signal in fractional Fourier domain. c): The ECG signal through 2D-LMS filter in fractional Fourier domain.

in fractional Fourier domain, Wiener filter and 2D-LMS filter are used to denoise $F_p(u)$ to obtain $F_p^*(u)$, where fractional Fourier transform of the clean signal $d(t)$ is also required. In the implementation of the code, the order of 2D-LMS filter is 2, and the iteration step is 0.01. Note that LMS filter has a lag effect, which leads the output of N-order LMS filter is N points less than the input. And for the inverse transformation in fractional domain, the length of output can make a big difference to the result, so it should make the length of output equal to the length of input. It just needs to append the first N values of the input to the output to obtain $F_p^*(u)$. Finally, do a inverse fractional Fourier transform (IFRFT) to get the denoised ECG signal $y(t)$ in time domain.

Figure 1 is a diagram of the data flow corresponding to the above steps.

4 SIMULATION EXPERIMENTS

4.1 Database

The selected database is Noise Stress Test Database (NSTDB) of MIT-BIH. This database includes 12 half-hour ECG recordings and 3 half-hour recordings of noise typical in ambulatory ECG recordings. Noise (BW/MA/EM) was added beginning after the first 5 minutes

of each record, during two-minute segments alternating with two-minute clean segments. In this paper, the data selected is the 5th-7th minute segment with a total of 7200 data points.

4.2 Experimental Results

The data used in the display of the legend is the 5th-7th minute segment of 118e12 in NSTDB, it indicates that the SNR of entire sample is 12db after adding noise. In Figure 2, a) is the clean ECG signal without noise, b) is the noisy ECG signal with noise, and c) is the denoised ECG signal in time domain. In Figure 3, a) is the selection process of the best FRFT order, b) is the ECG signal in fractional Fourier domain, and c) is the ECG signal through 2D-LMS filter in fractional Fourier domain.

In a) and b) of Figure 2, the change of the ECG signal after adding noise is obvious. By comparing a), b) and c), it can be found that after denoised by 2D-LMS filter in fractional Fourier domain, the influence of noise in the ECG signal of c) is obviously smaller than that of b), especially in terms of overall ups and downs.

Figure 4 enlarges the perspective of the first 1000 points of a), b) and c) in Figure 2, it can be found that not all the noise is removed in the details of every single heartbeat, but on the other hand, the method in this paper has a certain effect in terms of SNR and MSE.

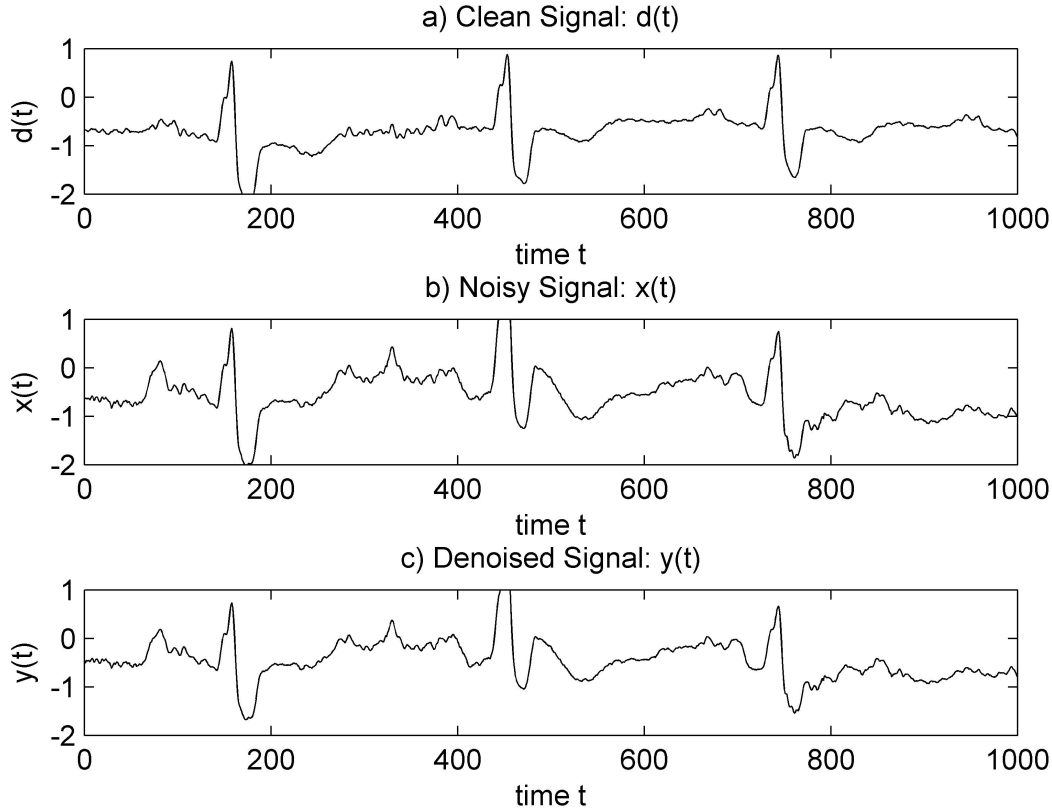


Figure 4: a): The clean ECG signal without noise. b): The noisy ECG signal with noise. c): The denoised ECG signal in time domain.

Table 1: Results comparison

	SNR (db)			MSE		
	x(t)	Wiener	2D-LMS	x(t)	Wiener	2D-LMS
data						
118e12	7.2827	5.5816	8.6099	0.1765	0.2611	0.1300
118e06	1.2625	3.6457	-27.5998	0.7058	0.4077	5.4317
118e18	13.2970	6.3840	11.1424	0.0442	0.2171	0.0725
118e24	19.3747	6.5313	12.8449	0.0109	0.2098	0.0490

In Table 1, the first column indicates the same data sample with different SNR noises, for example, 118e12 indicates that the SNR of entire sample is 12db after adding noise to the clean signal, the value of the second column is less than 12 because this paper only selects the 5th-7th minute segment. From Table 1, It can be found that the ECG denoising method based on Wiener filter in fractional Fourier domain performs well in terms of SNR, but the MSE is increased. This is because Wiener filter removed some low-frequency segment noise, but it also removed some of the original high-frequency segment of the ECG signal. The method of ECG signal denoising based on 2D-LMS filter in fractional Fourier domain has better

results in 118e12, 118e18, and 118e24, both in terms of SNR and MSE. However, the effect in 118e06 is relatively poor. This is because the SNR of the selected signal segment is 1.2625db, which means that the noise added to the data is particularly large. In this case, using the method in fractional Fourier domain to denoise the ECG signal needs to increase the number of sampling points to increase the SNR of the entire selected segment. In general, the performance of 2D-LMS filter-based denoising method in fractional Fourier domain for ECG data is still worth studying.

5 CONCLUSIONS

Based on the time-frequency transformation of signal, this paper transformed the ECG signal into the fractional Fourier domain for denoising, and proposed an improved 2D-LMS filter, which could be applied to the ECG signal in fractional Fourier domain. A complete set of method steps and a data flow diagram for denoising ECG signals with 2D-LMS filter in fractional Fourier domain were given. Simulation experiments were carried out with the data of MIT-BIH, using Wiener filter and 2D-LMS filter to denoise the ECG signal in fractional Fourier domain.

The results showed that the method based on the 2D-LMS filter in fractional Fourier domain has a relatively good effect when dealing with noisy signals with a SNR of 7db or more. But in some other cases, this method cannot get the correct signal in time domain and the effect is related to the number of sampling points. How to find a method in fractional Fourier domain with more stable and powerful effect is also the content to be concerned in the future work.

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